

The City of Numeria

Outbreak Simulation Challenge: A numerical adventure

Emergency broadcast. At the beginning of winter, hospitals in the city of Numeria report the first cases of a new disease, known locally as *Null Fever*, a disease that spreads quietly at first, then accelerates when the city reacts too late. The city is in a difficult situation, as it is isolated from the world during winter and can only access to outside aid after 6 month, the end of winter. The city council want to know:

If the hospital system can survive the next 180 days?

and what is the best policy to fight this outbreak?

The oracles of the city received a divine revelation regarding how the disease evolve, a set of differential equations! But they can not solve them, so your scientific team has been asked to build and use an adequate simulator. The simulator will not make decisions for you. It will sometimes mislead you if you use it carelessly. Your task is to understand both the epidemic and the numerical method.

The goal

By the end of the challenge, you should be able to answer questions such as:

- How large is the outbreak in the baseline scenario?
- When does the infectious population peak?
- Does the hospital load cross the danger line?
- Can a careless simulation create a fake epidemic curve?
- Which numerical method is more reliable: Euler, Heun, or fourth-order Runge–Kutta?
- Which intervention strategy gives the best result under limited resources?

The city dashboard

During the missions, track the following quantities:

$$\begin{aligned}
 \text{total population} \quad N(t) &= S(t) + V(t) + E(t) + I(t) + Q(t) + R(t), \\
 \text{hospital load} \quad H(t) &= 0.04I(t) + 0.10Q(t), \\
 \text{hospital capacity} \quad H_{\max} &= 600, \\
 \text{days over capacity} \quad D_{\text{over}} &= \int_0^{180} \mathbf{1}_{H(t) > H_{\max}} dt.
 \end{aligned}$$

The number $H(t)$ is not the number of infected people. It is an estimate of the occupied hospital beds: infectious people outside isolation have a smaller hospitalization fraction than isolated infectious people.

You will also track cumulative infections. Define $C(t)$ by

$$\frac{dC}{dt} = \lambda S + (1 - \eta)\lambda V,$$

with

$$C(0) = E(0) + I(0) + Q(0) + R(0).$$

Thus, $C(t)$ counts all people infected up to time t , including the initially infected people.

The epidemic model

The city population is divided into six groups:

Symbol	Name	Meaning
$S(t)$	Susceptible	can become infected
$V(t)$	Vaccinated	protected, but not perfectly
$E(t)$	Exposed	infected, not yet infectious
$I(t)$	Infectious	can infect others
$Q(t)$	Isolated	infectious, but with reduced contacts
$R(t)$	Recovered	immune for a while

The total population is

$$N = S + V + E + I + Q + R.$$

The effective transmission rate

Contacts are not constant as weekday rhythms, news reports, fear, and caution all change how often people meet. We model this by

$$\beta_{\text{eff}}(t, I) = \frac{\beta_0 \left(1 + a \sin\left(\frac{2\pi t}{7}\right)\right)}{1 + \kappa I/N}.$$

The factor

$$1 + a \sin\left(\frac{2\pi t}{7}\right)$$

adds a weekly rhythm. The denominator

$$1 + \kappa I/N$$

models behavioral feedback: when many people are infectious, citizens reduce contact.

The force of infection is

$$\lambda(t) = \beta_{\text{eff}}(t, I) \frac{I + \rho Q}{N}.$$

The parameter ρ measures how infectious isolated people still are. Since isolated people have fewer contacts, ρ is small.

The equations

The outbreak evolves according to

$$\begin{aligned}\frac{dS}{dt} &= -\lambda S - \nu S + \omega R, \\ \frac{dV}{dt} &= \nu S - (1 - \eta)\lambda V, \\ \frac{dE}{dt} &= \lambda S + (1 - \eta)\lambda V - \sigma E, \\ \frac{dI}{dt} &= \sigma E - \gamma I - \delta I, \\ \frac{dQ}{dt} &= \delta I - \gamma_Q Q, \\ \frac{dR}{dt} &= \gamma I + \gamma_Q Q - \omega R, \\ \frac{dC}{dt} &= \lambda S + (1 - \eta)\lambda V.\end{aligned}$$

The initial condition is

$$\begin{aligned} S(0) &= 99940, & V(0) &= 0, & E(0) &= 40, \\ I(0) &= 20, & Q(0) &= 0, & R(0) &= 0, \end{aligned}$$

and

$$C(0) = 60.$$

Simulate the city for

$$0 \leq t \leq 180$$

days.

Baseline parameters

Parameter	Value	Meaning
N	100000	total population
β_0	0.75	baseline transmission rate
a	0.10	strength of weekly contact rhythm
κ	25	strength of behavioral feedback
ρ	0.05	infectiousness of isolated people
η	0.80	vaccine protection
ν	0.0005	vaccination rate per day
ω	1/240	immunity waning rate
σ	1/4	exposed-to-infectious rate
γ	1/8	recovery rate from infectious state
δ	0.08	isolation rate
γ_Q	1/10	recovery rate from isolation

All rates are measured per day.

Mission 0: Study the model

Before the city trusts your simulator, it wants to know whether the equations at least obey basic logic. Does the model conserve population? What does each term mean?

Tasks

1. Identify the terms that move people out of S .
2. Identify the terms that move people into E .
3. Identify the terms that move people from I to Q .
4. Identify the terms that move people back from R to S .
5. Add the six equations for S, V, E, I, Q, R and simplify.
6. Verify that

$$\frac{d}{dt}(S + V + E + I + Q + R) = 0.$$

What does this say about the city population?

Mission 1: Build the first forecast

The council asks for a first 180-day forecast. The simplest emergency tool is explicit Euler. It is fast, easy to implement, and dangerous if used blindly.

Let

$$Y = (S, V, E, I, Q, R, C).$$

Write the system as

$$\frac{dY}{dt} = F(t, Y).$$

The explicit Euler method is

$$Y_{n+1} = Y_n + \Delta t F(t_n, Y_n),$$

with

$$t_{n+1} = t_n + \Delta t.$$

Use

$$\Delta t = 0.25$$

days.

Tasks

1. Simulate the model from day 0 to day 180.
2. Plot $E(t)$, $I(t)$, and $Q(t)$ on one graph.
3. Plot $S(t)$, $V(t)$, and $R(t)$ on another graph.
4. Plot $H(t)$ and the horizontal danger line $H_{\max} = 600$.
5. Plot $C(t)$.

Questions.

1. On which day does $I(t)$ reach its maximum?
2. What is the maximum value of $I(t)$?
3. On which day does the hospital load $H(t)$ reach its maximum?
4. How many days does the city spend above hospital capacity?
5. What is the final value of $C(t)$?
6. Does the total population $N(t)$ remain close to 100000?
7. Do all compartments remain nonnegative?

Mission 2: The simulation can lie

The first forecast looks official, but the city scientist is suspicious. A numerical method is not the same as the mathematical model. To test the simulator, deliberately stress it.

Repeat Mission 1 using explicit Euler with

$$\Delta t = 0.5, \quad 1, \quad 2, \quad 4, \quad 8.$$

For each run, record:

- the peak value of I ;
- the day of the peak;
- the peak hospital load;
- the minimum value among S, V, E, I, Q, R ;
- the maximum population error

$$\max_t |N(t) - 100000|;$$

- whether the plot looks physically plausible.

Questions.

1. At which time step does the simulation first become suspicious?
2. At which time step does it clearly become impossible?
3. Does population conservation guarantee that the numerical result is correct?
4. Can you find a run where $N(t)$ stays almost constant but one compartment becomes negative?

Mission 3: The quarantine shock

A new emergency protocol is announced. Exposed people are detected quickly, infectious people are isolated quickly, and isolated people recover quickly. Biologically this seems good. Numerically, it creates very fast time scales.

Change only these parameters:

$$\delta = 2.5, \quad \gamma_Q = 1.0, \quad \sigma = 1.5.$$

Run explicit Euler with

$$\Delta t = 0.05, \quad 0.1, \quad 0.25, \quad 0.5, \quad 1.0.$$

Questions.

1. Which time steps produce reasonable curves?
2. Which time steps create oscillations or negative compartments?
3. This situation is often called stiffness. Based on your experiments, what does stiffness feel like computationally?

Mission 4: Three engines for the same city

Euler was quick but fragile. The city offers you two upgraded engines: Heun's method and fourth-order Runge–Kutta. You must decide whether the upgrades are worth it.

Engine 1: Euler

$$Y_{n+1} = Y_n + \Delta t F(t_n, Y_n).$$

Engine 2: Heun's method

$$\begin{aligned} K_1 &= F(t_n, Y_n), \\ K_2 &= F(t_n + \Delta t, Y_n + \Delta t K_1), \\ Y_{n+1} &= Y_n + \frac{\Delta t}{2} (K_1 + K_2). \end{aligned}$$

Engine 3: Fourth-order Runge–Kutta

$$\begin{aligned} K_1 &= F(t_n, Y_n), \\ K_2 &= F\left(t_n + \frac{\Delta t}{2}, Y_n + \frac{\Delta t}{2} K_1\right), \\ K_3 &= F\left(t_n + \frac{\Delta t}{2}, Y_n + \frac{\Delta t}{2} K_2\right), \\ K_4 &= F(t_n + \Delta t, Y_n + \Delta t K_3), \\ Y_{n+1} &= Y_n + \frac{\Delta t}{6} (K_1 + 2K_2 + 2K_3 + K_4). \end{aligned}$$

Comparison task

Use a very accurate run as a benchmark. For example, use RK4 with

$$\Delta t_{\text{ref}} = 0.01$$

or smaller.

Compare Euler, Heun, and RK4 using

$$\Delta t = 2, \quad 1, \quad 0.5, \quad 0.25.$$

At the same output times, compute

$$e_I(\Delta t) = \max_j |I_{\Delta t}(t_j) - I_{\text{ref}}(t_j)|.$$

You may also compute a full-state relative error,

$$e_Y(\Delta t) = \max_j \frac{\|Y_{\Delta t}(t_j) - Y_{\text{ref}}(t_j)\|_{\infty}}{100000}.$$

Questions.

1. For the same Δt , which method is most accurate?
2. Which method gives the best peak day and peak size?
3. Does RK4 always stay physically meaningful when Δt is very large?
4. Which is more important in your experiment: higher order, smaller time step, or both?

Mission 5: Measure the order of convergence

A method is not only judged by whether it looks good once. A numerical scientist asks a sharper question: when the time step is divided by two, how fast does the error decrease?

For a method of order p , one expects approximately

$$e(\Delta t) \approx C(\Delta t)^p$$

when Δt is small enough and the solution is smooth enough.

Taking logarithms gives

$$\log e(\Delta t) \approx \log C + p \log(\Delta t).$$

So the slope of a log-log plot estimates p .

Tasks

For each method, use

$$\Delta t = 1, \quad \frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}.$$

Use a high-accuracy RK4 solution as the benchmark and compute $e_I(\Delta t)$.

Then:

1. Plot $e_I(\Delta t)$ against Δt on a log-log plot.
2. Estimate the observed order using

$$p_{\text{obs}}(\Delta t) = \frac{\log(e_I(\Delta t)/e_I(\Delta t/2))}{\log 2}.$$

3. Repeat using the full-state error $e_Y(\Delta t)$ if time allows.

Questions.

1. Does Euler behave like a first-order method?
2. Does Heun behave like a second-order method?
3. Does RK4 behave like a fourth-order method?
4. What happens if you include a time step that is too large and unstable?
5. What happens if the error becomes extremely small? Can roundoff error disturb the slope?

Mission 6: Intervention cards

The council has to decide an adequate intervention. Each intervention has a cost. Your team has five action points. Spend them wisely.

The baseline parameters remain active unless changed by an intervention.

Intervention card	Cost	Effect
Public caution campaign	2	Increase κ by 15
Faster vaccination	3	Set $\nu = 0.0015$
High-protection booster	2	Set $\eta = 0.90$
Rapid testing and isolation	2	Set $\delta = 0.16$
Safer isolation centers	1	Set $\rho = 0.02$
Festival crowd limits	3	For days $0 \leq t \leq 30$, replace β_0 by $0.80\beta_0$

You may spend at most five action points. If two cards modify the same parameter, use the stronger effect rather than stacking them.

Scoring

A good strategy should:

- keep $H(t)$ below 600 if possible;
- reduce the number of days over capacity;
- reduce cumulative infections $C(180)$;
- avoid relying on a numerically unstable time step.

Questions.

1. Which combination of cards gives the lowest peak hospital load?
2. Which combination gives the lowest cumulative infections?
3. Are these the same strategy?
4. Does an early temporary reduction in β_0 merely delay the peak, or does it reduce it?
5. Can two moderate interventions combine into a surprisingly strong result?

Mission 7: The hidden outbreak parameters

The first forecasts were based on assumed parameters. Now surveillance teams send imperfect weekly estimates of the infectious population. The council asks a harder question: what transmission parameters best explain the data?

Assume all baseline parameters are known except β_0 and κ . You receive the following weekly observations:

Day	7	14	21	28	35	42	49
Observed I	110	440	1980	4400	7440	8250	7370
Day	56	63	70	77	84	91	
Observed I	7230	5380	4680	3250	2990	2400	

The observations are noisy. Do not expect a perfect match.

Parameter search

Try a grid such as

$$0.60 \leq \beta_0 \leq 0.95, \quad 5 \leq \kappa \leq 45.$$

For each pair (β_0, κ) , simulate the model and compare the simulated infectious values to the observations. One useful loss is

$$L(\beta_0, \kappa) = \frac{1}{m} \sum_{j=1}^m [\log(1 + I_{\text{model}}(t_j)) - \log(1 + I_{\text{obs}}(t_j))]^2.$$

The logarithm prevents the largest observations from completely dominating the fit.

Tasks

1. Find the pair (β_0, κ) with the smallest loss.
2. Plot the observed data and the best-fitting simulated curve.
3. Plot $L(\beta_0, \kappa)$.
4. Find several parameter pairs that fit almost as well as the best one.

Questions.

1. Is there one sharp best parameter pair, or a valley of nearly good pairs?
2. Can a larger β_0 be partially compensated by a larger κ ?
3. If you only had the first five weeks of data, would your estimate change?
4. How would wrong parameters affect the intervention chosen in Mission 6?

Mission 8: Final mission: robust decisions under uncertainty

The council is no longer satisfied with a single forecast. A single best-fit parameter set may be a trap. Several parameter pairs can explain the data almost equally well, but they may predict different futures. Numeria needs a policy that survives uncertainty.

Use the parameter search from Mission 7. Let

$$L_{\min} = \min_{\beta_0, \kappa} L(\beta_0, \kappa).$$

Build a small uncertainty set, for example

$$\mathcal{P} = \{(\beta_0, \kappa) : L(\beta_0, \kappa) \leq L_{\min} + 0.02\}.$$

If this set is too large, keep a representative sample of parameter pairs: some with low β_0 , some with high β_0 , some with low κ , and some with high κ .

Now return to the intervention cards. For each possible policy, simulate every parameter pair in $\mathcal{P}$.

Define the worst-case hospital load

$$H_{\text{worst}} = \max_{(\beta_0, \kappa) \in \mathcal{P}} \max_{0 \leq t \leq 180} H(t),$$

and the worst-case days over capacity

$$D_{\text{worst}} = \max_{(\beta_0, \kappa) \in \mathcal{P}} D_{\text{over}}.$$

Tasks

1. Choose a policy with at most five action points.
2. Evaluate it over all parameter pairs in $\mathcal{P}$.
3. Compare it with the policy that looked best using only the single best-fit parameter pair.
4. Decide whether the city should optimize for the average case or the worst case.

Questions.

1. Which policy is best in the worst case?
2. Is it the same as the policy chosen from the baseline model?
3. What does this tell you about forecasting under uncertainty?
4. If the city could buy one extra action point, which intervention would become attractive?
5. What additional data would most reduce the uncertainty: more infection reports, hospital reports, vaccination data, contact surveys, or isolation compliance data?